

Sepehr Khavari

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EDUCATION

Northwestern University McCormick School of Engineering Evanston, IL
MS (Mechanical Engineering) – *expected June 2027*

- Research focus: Medical Device Development, Neurosurgery, Biomechanics.

Northwestern University Weinberg College of Arts and Sciences Evanston, IL
BA (Neuroscience) – **GPA: 4.00/4.00** – *expected June 2027*

- Honors: Dean's List, QuestBridge Scholar, Arch Scholar

Plano West High School Plano, TX
Diploma – *June 2024*

- Honors: **Summa Cum Laude**, TMEA Math UIL Area Finalist, Top 1% in Class.
- *SAT – 1580/1600, ACT – 35/36*

MEDICAL DEVICE PATENTS & COMPANIES

1. ReView ocular flushing system for chronic allergic keratoconjunctivitis and related conditions
 - a. **Provisional Patent**, Northwestern University INVO
 - i. Contributor in all aspects, including conceptualization, prototype development, 3D-printing, synthetic model testing, funding and grant writing, collaborations, regulatory, and interfacing with incubators and venture capital firms.
2. FASTEN8 intelligent pedicle screw system to detect instrumentation failure after spinal surgery
 - a. **Provisional Patent**, Northwestern University INVO
 - i. Selected for the **InQbation** medical device development program at the **Kellogg School of Business**.
 - ii. Contributor in all aspects, including conceptualization, prototype development, synthetic model testing, porcine animal testing, funding and grant writing, collaborations, interfacing with incubators and venture capital firms, regulatory, etc.
3. MRI-compatible spinal cord stem cell injection system
 - a. **Provisional Patent**, Northwestern University INVO
 - i. Contributor in CAD design, iterating, derisking, and commercialization.
4. Bionic prosthetic penis
 - a. **Provisional Patent**, Northwestern University INVO
 - i. Contributor in all aspects, including conceptualization, prototype development, synthetic model testing, funding and grant writing, and collaborations.

RESEARCH EXPERIENCE

Functional Research Group, Department of Neurosurgery (Northwestern Feinberg) Chicago, IL
Researcher; PIs: Joshua Rosenow, MD; Christopher Ahuja, MD, PhD Dec 2024 – Present

- Involved with a team of 15 medical students collaborating on projects in functional and spine neurosurgery.
- Areas of focus: Neuromodulation, Focused Ultrasound, Spinal Implants, Neurosurgical Robotics, SCS.
- Individual Media Recognition
 - o <https://dailynorthwestern.com/2025/10/14/campus/northwestern-sophomore-to-present-two-research-posters-at-prestigious-neuroscience-conference/#>

John Rogers Lab at Querrey-Simpson Institute for Bioelectronics (Northwestern University) Chicago, IL
Researcher; PI: John Rogers, PhD Jan 2025 – Present

- Developing a novel medical device to enable real-time detection of pedicle screw integrity after spinal surgery.
- Validating a novel diagnostic microphone system for detecting failure in spinal fusion instrumentation.

SKILLS & INTERESTS

Personal Interests: LEGO® building and collecting, avid horology enthusiast and small-time collector, lover of Star Wars

Academic Interests: Functional neurosurgery, ultrasound neuromodulation, medical device development, neuroethics, and biomechanical gait analysis.

Skills: Proficient in CAD (SolidWorks and AutoCAD), Python, PyTorch, MATLAB, basic lab science techniques, and electronic and PCB design. Bambu Studio 3D-Printing Certification (Bambu Lab). Rat Tickling Certification (Purdue University). Native Farsi speaker

HONORS & AWARDS

NEURON Scholar – Nov 2025 – Present

Northwestern University Department of Neurobiology

Arch Scholar – June 2024 – Present

Northwestern University

Dean's List – Sep 2024 – Present

Northwestern University

QuestBridge Scholar (Full-Ride Scholarship) – Dec 2023 – Present

Northwestern University

Top 100 Entries in Medical Devices Category – Sep 2024

Create the Future Design Contest – SAE Media Group

2024 STEM Signing Day Award – May 2024

Boeing Co

Music Changing Lives Scholarship – Apr 2024

Lone Star Youth Winds

Rise for the World Global Finalist – Oct 2023

Rhodes Trust & Eric and Wendy Schmidt Fund for Strategic Innovation

ORAL & POSTER PRESENTATIONS

1. Low-Intensity Focused Ultrasound Neuromodulation for Dementia: A Scoping Review, **CNS 2025**, Los Angeles, CA.
2. Invasive Vagus Nerve Stimulation for Non-Epilepsy Indications: A Scoping Review, **CNS 2025**, Electronic Poster, Los Angeles, CA
3. Early Detection of Pedicle Screw Loosening Using a Tri-Axial Smart Fixation System, **Chicago Area Undergraduate Research Symposium (CAURS) 2026**, Oral Presentation, Chicago, IL.
4. Biologic Augmentation of Orthopedic Implants with Stem Cells: A Scoping Review of Clinical Advancements **Chicago Spine Research Symposium 2025**, Chicago, IL.
5. Assessing the Utility of Pedometer Data in Predicting Spine Surgery Outcomes: A Scoping Review, **NANS 2026**, Electronic Poster, Las Vegas, Nevada
6. Early Detection of Pedicle Screw Loosening Using a Tri-Axial Smart Fixation System, **Chicago Society for Neuroscience 2026**, Printed Poster Chicago, IL.

SELECT PUBLICATIONS

1. Kaniuk J.K., Kumar D., Mazurek C., **Khavari S.**, Sollenberger C., Ahuja A., Mossner J.M., & Ahuja C. From Acute Injury to Chronic Neurodegeneration: Molecular Mechanisms Linking Secondary Brain Injury to Long-Term Pathology. *International Journal of Molecular Sciences*. 2025;26(15):7191. doi:10.3390/ijms26157191
2. Mittal M., D'ambrose N., Jain R., Samal B., **Khavari S.**, Prasad D., Ramanos S., & Sharan A. ERAS protocols: Impact on outcomes in spine surgery. *Seminars in Spine Surgery*. 2025;37(3):101188. doi:10.1016/j.semss.2025.101188

Mentors: Christopher Ahuja, Joshua Rosenow, Dillan Prasad, Yue Wang, Rishi Jain